

# LAURA RINALDI

---

Piove di Sacco (PD), 27/07/1996

e-mail: [laura.rinaldi96@libero.it](mailto:laura.rinaldi96@libero.it)

phone: [REDACTED]

ORCID: 0000-0002-7999-6538

Website: <https://laura-rinaldi.github.io/>

GitHub: <https://github.com/laura-rinaldi>

## RESEARCH POSITIONS

---

2025 - today      **Junior Researcher** at the Institute of Applied Mathematics and Information Technologies "E. Magenes" (IMATI), Pavia, contributing to the project "Digital Driven Diagnostics, prognostics and therapeutics for sustainable Health care" D34 Health – PNC 0000001 - CUP B53C22006100001 – Spoke 3.

2025, 11 mos.      **Postdoctoral researcher** at the University of Padua, contributing to the project "Tchakaloff-Davis-Wilhelmsen measure compression and applications to numerical modelling".

## EDUCATION

---

May 2025      Participating in Virtual-exchange course: "Learning-enhanced Data Assimilation" (4 CFU), promoted by the University of Padova and the University of Southern California (USA).

31-03-2025      **PhD in Mathematical Sciences** at the University of Padua, with a doctoral thesis on mathematical physics, modeling, numerical analysis, and machine learning. The thesis, titled "*A mathematical construction of the digital twin of bread leavening*", was developed within the framework of the project "Reduction of energy waste in food and beverage preparation activities". SOCINT-G-Research Award for the **best PhD thesis** in quantitative disciplines.

Jul 2024      Participating in Udine CISM "Advanced Course on Inverse Problems for Mechanical Systems: Methods, Simulations and Experiments".

Jul 2023      Participating in the "Barcelona XX Jacques-Louis Lions Spanish-French School on Numerical Simulations in Physics Engineering". **Best poster** presented "Bread leavening and baking: a modelling challenge".

25-09-2020      **Master's Degree in Mathematics** at the University of Padua with 110/110 with honors and thesis in the physical-mathematical field "*Bose Hubbard model of cold atoms to higher orders*".

28-09-2018      **Bachelor's degree in Mathematics** at the University of Padua with 100/110 and thesis in the physical-mathematical field entitled "*Shading properties and topological stability for discrete dynamical systems*".

2015      High School Diploma at "Istituto d'istruzione superiore A. Einstein", Scientific High School - Piove di Sacco.

## PROFESSIONAL WORK EXPERIENCES

---

2021, 10 mos.      Apprenticeship at the "SCAI ITEC S.r.l." company in Padua with database administration duties.

2020, 4 mos.      Internship at the company "SCAI ITEC S.r.l." in Padua with training objectives in database administration.

2020, 1 mo.      "Ready to Oracle" training course to learn Oracle Database, SQL Server and, SQL Language.

30-03-2021      Oracle Database SQL Certified Associate 1Z0-071.

## TEACHING ACTIVITIES

---

|              |                                                                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 - 2026  | <b>Teaching</b> of “Istituzioni di matematica con esercitazioni – mod.1” for the degree course in Environmental Science at the University of Ca’ Foscari (Venice). Teacher (48 h).                                                                      |
| 2025, 4 mos. | <b>Supplementary teaching activity</b> of “Calcolo Numerico” for the degree course in Energy Engineering, Prof. Alvise Sommariva at the University of Padua. Teacher support activities during laboratory with Matlab exercises (24 h).                 |
| 2022, 4 mos. | <b>Supplementary teaching activity</b> of “Calcolo Numerico” for the degree course in Mathematics, Prof. Michela Redivo-Zaglia at the University of Padua. Teacher support during laboratory with Python/Matlab scripts and exercises (25 h).           |
| 2021, 4 mos. | <b>Supplementary teaching activity</b> of “Fondamenti di Analisi Matematica 2” for the degree course of Chemical and Materials Engineering (nr. Channel 1), Prof. Luca Baracco at the University of Padua. Exercises and solution of past exams (24 h). |

## MENTORING EXPERIENCE

---

|            |                                                                                                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19-09-2025 | Co-supervisor of M. Storgato. <b>Bachelor’s thesis</b> “ <i>Quadratura subperiodica in python</i> ”.                                                                            |
| 19-09-2025 | Co-supervisor of G. Traversin. <b>Bachelor’s thesis</b> “ <i>Implementazione in Python di un algoritmo per la cubatura numerica su elementi poligonali con un lato curvo</i> ”. |
| 11-04-2025 | Co-supervisor of E. Caregnato. <b>Master’s thesis</b> “ <i>Goal Oriented Operator Networks</i> ”.                                                                               |
| 19-04-2024 | Co-supervisor of M. Dell’Orto. <b>Master’s thesis</b> “ <i>Variationally mimetic operator neural networks for the time-dependent heat equation</i> ”.                           |

## AWARDS AND PRIZES

---

|      |                                                                                                                                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | Winner of the “SOCINT- G-Research Award” for <b>the best PhD thesis</b> in quantitative subjects – II Edition.                                    |
| 2024 | <b>Third place</b> in the “Three-Minutes Thesis Competition (3MT)” at the University of Padua, promoted by Coimbra Group.                         |
| 2023 | Winner of the <b>best poster</b> presented at the “XX Jacques-Louis Lions Spanish-French School on Numerical Simulations in Physics Engineering”. |
| 2023 | Winner of the <b>best presentation</b> among PhD students at the conference “FEniCS 2023”.                                                        |

## GRANTS AND SCHOLARSHIPS

---

|           |                                                                                                                                     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------|
| 2026      | GNCS-INDAM Project “Metodi computazionali avanzati per sistemi complessi: approcci ibridi numerici e machine learning” (1700,00 €). |
| 2023      | GNCS-INDAM grant for participating in the “SIMAI 2023” conference in Matera.                                                        |
| 2018/2019 | Regional scholarship winner.                                                                                                        |
| 2018      | Grant for participating in the “Junior Math Days” workshop at the SISSA University of Trieste.                                      |
| 2017/2018 | Regional scholarship winner.                                                                                                        |
| 2016/2017 | Regional scholarship winner.                                                                                                        |

## AFFILIATIONS TO MATHEMATICAL GROUPS AND PROFESSIONAL SOCIETIES

---

|            |                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------|
| Since 2023 | Member of the “Società Italiana di Matematica Applicata e Industriale” (SIMAI).                                          |
| Since 2023 | Early Career Membership of the “Society of Industrial and Applied Mathematics” (SIAM). Before 2026 (student membership). |
| Since 2022 | Member of the “Gruppo Nazionale Calcolo Scientifico” (GNCS) of the Istituto Nazionale di Alta Matematica (INdAM).        |
| 2022       | Member of “Unione Matematica Italiana” (UMI).                                                                            |

## CONFERENCES AND SEMINARS

---

|          |                                                                                                                                                                                           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mar 2026 | CNR. Seminari di Matematica Applicata.<br><u>Talk</u> “ <i>The role of surrogate models in different applied cases: bread leavening and fluid dynamics</i> ” – <b>invited</b> .           |
| Feb 2026 | Cagliari CSMM 2026.<br><u>Poster</u> “ <i>Operator network for solving Navier-Stokes equations</i> ”.                                                                                     |
| Sep 2025 | Trieste SIMAI 2025.<br><u>Talk</u> “ <i>Building a surrogate model for coupled, time-dependent PDEs using Operator Networks for thermal applications</i> ” – <b>invited</b> .             |
| Apr 2025 | University of Padua. Mathematical Physics and Dynamical Systems Seminar.<br><u>Talk</u> “ <i>A mathematical model of bread leavening for building a digital twin</i> ” – <b>invited</b> . |
| Aug 2023 | Matera SIMAI 2023.<br><u>Talk</u> “ <i>On approximating fictitious heat sources arising in an inverse heat transfer problem</i> ” – <b>invited</b> .                                      |
| Jun 2023 | Pula FEniCS 2023.<br><u>Talk</u> “ <i>Bread baking simulation with FEniCS</i> ”.                                                                                                          |
| Sep 2022 | Arenzano YAMC Second Conference of Young Applied Mathematicians.<br><u>Talk</u> “ <i>Modelling for bread preparation to avoid energy waste</i> ”.                                         |
| Sep 2022 | Pavia GIMCS SIMAI Young 2022.<br><u>Talk</u> “ <i>Modelling for bread preparation to avoid energy waste</i> ”.                                                                            |
| May 2020 | University of Padua.<br><u>Talk</u> “ <i>The quantum harmonic oscillator: analysis and resolution</i> ”.                                                                                  |

## WORKSHOPS

---

|          |                                                                                                                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| May 2025 | University of Pavia. 7th Lombardy Young Numerical Analysts Meeting (LYNUM-VII).                                                                                                                                |
| Feb 2025 | Padova DTW 2025 International Workshop on Digital Twins: Mathematical Analysis, Formal Methods and Scientific Machine Learning.<br><u>Poster</u> “ <i>Digital twin of bread leavening for energy saving</i> ”. |
| Nov 2024 | Milano M4E: mathematics for Planet Earth.<br><u>Poster</u> “ <i>Digital twin of bread leavening for energy saving</i> ”.                                                                                       |
| Dec 2018 | SISSA University of Trieste. Workshop “Junior Math Days”.                                                                                                                                                      |

## SCIENTIFIC PUBLICATIONS

---

### SUBMITTED

1. **L. Rinaldi**, G. G. Giusteri. "Variationally mimetic operator network approach to transient viscous flow"  
<https://arxiv.org/abs/2604.02124>
2. **L. Rinaldi**, A. Sommariva, M. Vianello. "ORTHOCUB: integral and differential cubature rules by orthogonal moments"  
<https://arxiv.org/abs/2512.06049>

### PUBLISHED

3. **L. Rinaldi**, A. Sommariva, M. Vianello. "Effective numerical integration on complex shaped elements by discrete signed measures", Finite Elements in Analysis and Design, Volume 254, (2026)  
<https://doi.org/10.1016/j.finel.2025.104505>
4. E. Chinellato, P. Martin, **L. Rinaldi**, F. Marcuzzi. "Exploiting scientific machine learning on embedded digital twins", Springer series: Lecture Notes in Computational Science and Engineering - Math to Product (2025),  
[https://link.springer.com/chapter/10.1007/978-3-031-95709-3\\_12](https://link.springer.com/chapter/10.1007/978-3-031-95709-3_12)
5. **L. Rinaldi**, G. G. Giusteri. "Modelling and simulation of bread leavening to monitor energy consumption in industrial processes", Commun. Appl. Ind. Math. 16(1), 41-61 (2025) <https://doi.org/10.2478/caim-2025-0003>
6. G. G. Giusteri, F. Marcuzzi, **L. Rinaldi**. "Replacing voids and localized parameter changes with fictitious forcing terms in boundary-value problems", Results Appl. Math. 20, 100402 (2023) <https://doi.org/10.1016/j.rinam.2023.100402>

### DOCTORAL THESIS

7. **L. Rinaldi**. "A mathematical construction of the digital twin of bread leavening". Ph.D. Thesis (2025)  
<https://tesidottorato.depositolegale.it/handle/20.500.14242/218031>

## LANGUAGES

---

**Italian:** mother tongue.

**English:** good knowledge of spoken and written language, B2 certified at the University Language Center (CLA) of the University of Padua.

**French:** school level.

## COMPUTER SKILLS

---

**Coding:** Matlab/Octave, Python (particular, FeniCS/FeniCSX libraries for solving partial differential equations, PyTorch for deep learning), LATEX (Beamer for presentation), Wolfram Mathematica, SQL query language

**Database:** Oracle, SQL Server

**Other tools:** KNIME, SQL Developer, Toad, Putty, FileZilla, and WinSCP, Git/GitHub, Docker

**Software:** Paraview, Microsoft Office

**OS:** Linux/Ubuntu, Windows

## SOFTWARES

---

A code repository is available on my GitHub page: [laura-rinaldi \(Laura Rinaldi\) · GitHub](#)

## ORGANIZATION AND EVENTS

---

Since 2023      Organizer duties for the "Terza Missione" activity **Science4all**: a volunteering initiative promoted by the University of Padua, designed for primary school students aged 8 to 11. The program offers interactive workshops and hands-on science labs that aim to spark curiosity, foster critical thinking, and inspire a love for science in a fun and engaging way.

## EXTRACURRICULAR INTERESTS

---

Practiced at the amateur level in figure skating, speed skating, and ice skating.  
License of volleyball referee manager obtained at the provincial committee FIPAV of Padua.  
Certificate of attendance in the piano course.  
Practiced activities with the task of assistant-educator for children aged 3 to 6 years.

Autorizzo il trattamento dei miei dati personali presenti nel curriculum vitae ai sensi del Decreto Legislativo 30 giugno 2003, n. 196 e del GDPR (Regolamento UE 2016/679).

Padova, 07/04/2026

Laura Rinaldi